# API Manager Runtime



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WSO2

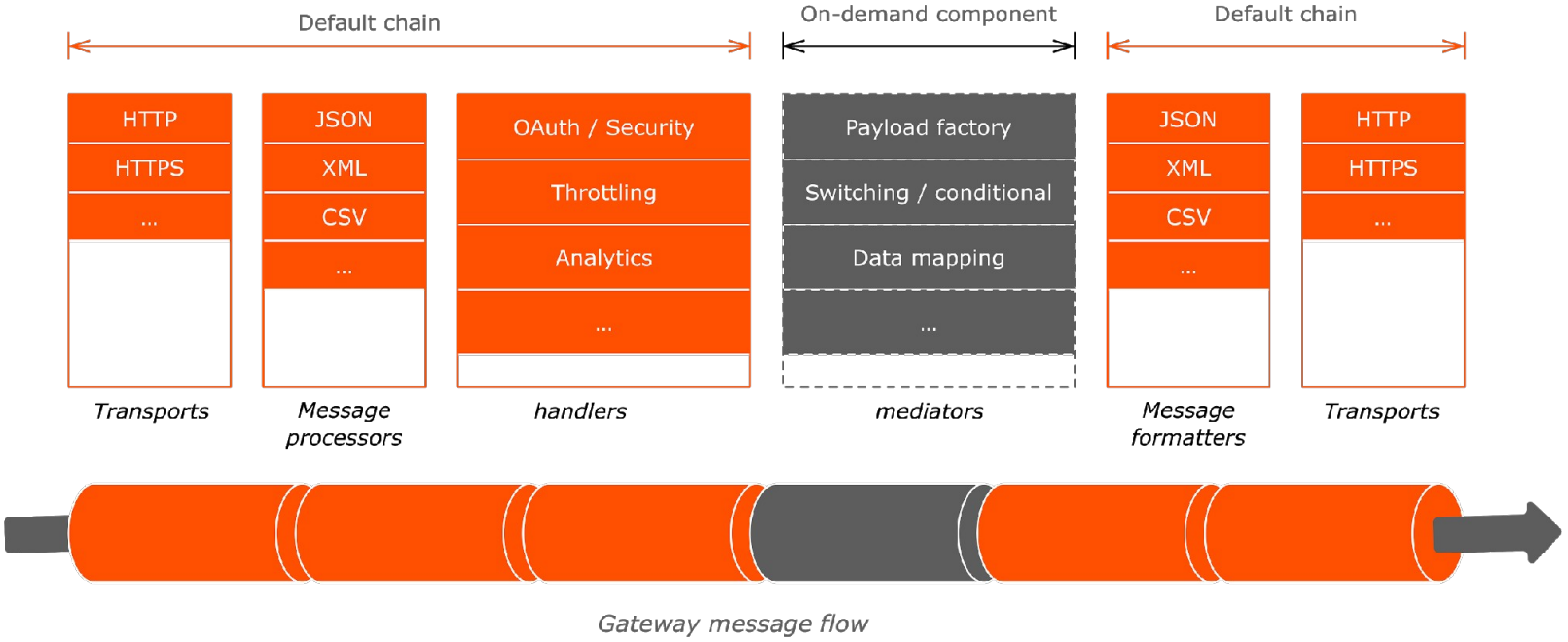
# API Gateway

WSO2

# WSO2 API Gateway

- WSO2 API Gateway (Classic), which is powered by Apache Synapse, is an API proxy which enforces policies for API requests.
- It secures and manages API calls by intercepting API requests and applying policies such as rate limiting, message mediation and security using handlers and managing API statistics.
- After validating all policies, the Gateway passes API calls to the actual back end.
- When the API Manager is running, you can access the Gateway via https://<GW_HOST>:8243 or http://<GW_HOST>:8280.

# API Gateway Architecture




*Gateway message flow*

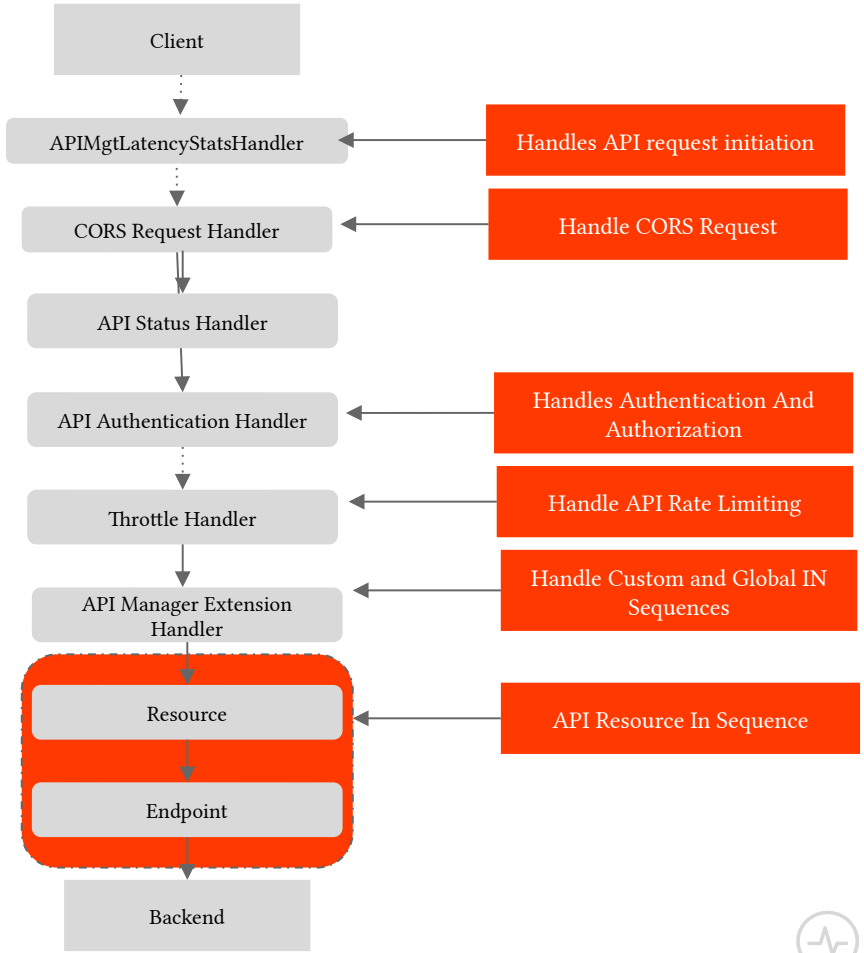
| Component | Usage |
| --- | --- |
| Transports | • When an API request first hits the WSO2 API Gateway it is received by the HTTP/HTTPS transports.<br>• The transport is responsible for carrying messages that are in a specific format.<br>• The transport provides a receiver, which is used to receive messages, and a sender, which is used to send messages. |
| Message Processors | • Then the receiving transport selects a message builder based on the message's content type.<br>• It uses the builder to process the message's raw payload data and convert it into a common XML, which the Gateway mediation engine can then read and understand.<br>• The Gateway includes message builders for text-based and binary content. |
| Handlers | • Then the request is passed through a set of handlers that applies the quality of services on the request message.<br>• Further, it enforces security, rate-limiting, and transformations if any on API requests while feeding valuable information of these requests to API Analytics. |

| Component | Usage |
| --- | --- |
| API Policies (Mediators) | • After the handlers, the message is passed to the mediators.<br>• Any custom mediation sequence will be applied as an API Policy. You can change the message type, set header, set payload, etc. |
| Message Formatters | • After mediators, the message is passed to the message formatters.<br>• Message formatters are used to convert the message type to proper format that needs to be sent to the backend.<br>• A message formatter is selected based on the message's content type. |

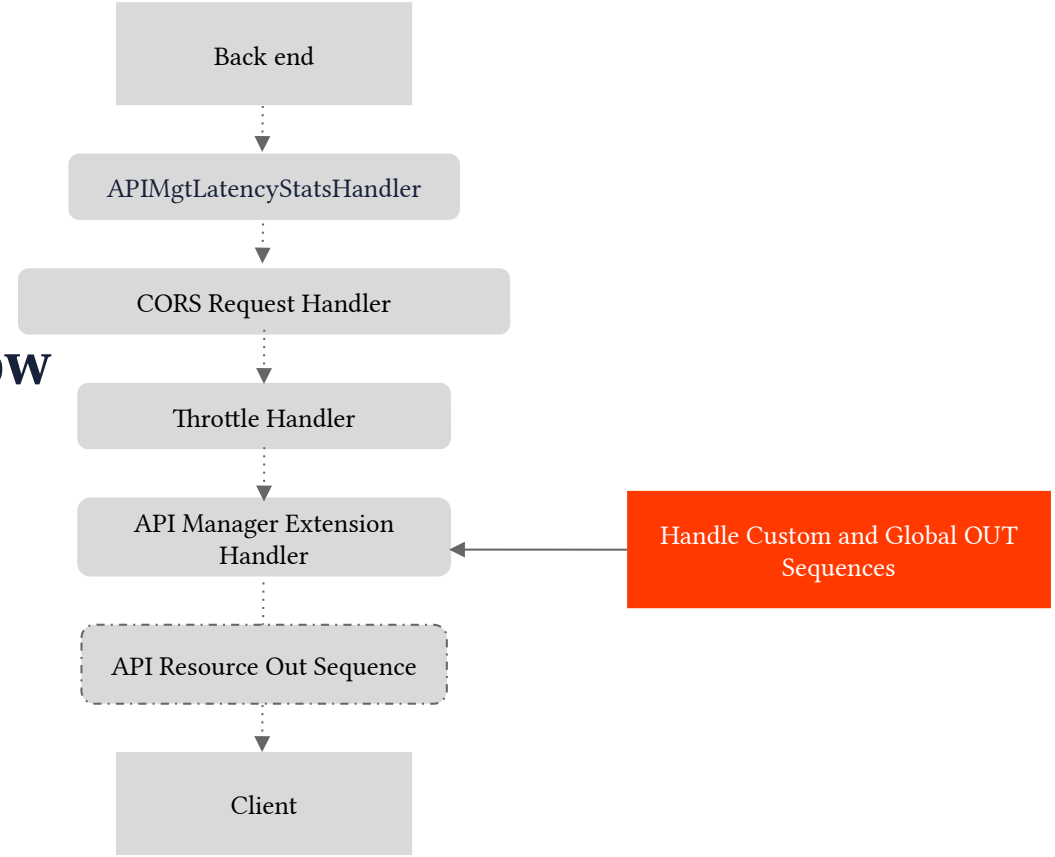
# WSO2 API Gateway

- In the end, the request is routed to the backend endpoint.
- Conversely, before transport sends a message out from the Gateway, a message formatter is used to build the outgoing stream from the message back into its original format.
- As with message builders, the message formatter is selected based on the message's content type.
- Message building and formatting only occurs if there is a content-aware mediator (i.e., reading the request payload) in the flow.

# Gateway Request Flow



# Gateway Response Flow



# Main Features of the API Gateway

The Gateway supports the below features to control access and enforce security.

- Supports multiple API types: REST, SOAP, Async APIs (WebSocket/SSE, WebSub), AI/LLM.
- Supports JWT, OAuth2.0, Basic Auth, API Key, Mutual TLS, and custom authentication schemes.
- Validate API payload contents against schemas.
- Apply additional security policies to APIs (authentication and authorization).
- Supports all standard OAuth2.0 grant types and allows extensions and additions to grants.
- Works seamlessly with third-party OAuth2.0 providers, standard, or proprietary.
- Allows blocking subscriptions due to non-payment, API abuse, etc.
- Associate API to system-defined service tiers for quotas and rate-limits.
- Generate JSON web tokens for consumption by back-end services.
- Threat protection, bot detection, and token-fraud detection.
- Supports detection of abnormal system use through artificial intelligence and machine learning.

# Gateway REST API

| | |
|---|---|
| POST /redeploy-api | Redeploy the API in the Gateway |
| POST /undeploy-api | Undeploy the API in the Gateway |
| GET /server-startup-healthcheck | Healthcheck to check if all the APIs are deployed during the server startup |
| GET /api-artifact | Get API synapse definition from the storage |
| GET /local-entry | Get Local Entry from the storage |
| GET /sequence | Get sequences from the storage |
| GET /end-points | Get endpoints from the storage for the API |
| GET /apis | Get the list of APIs by providing context and version |
| GET /apis/{apiId} | Get the subscription information of an API by providing the API uuid |
| GET /applications | Get the applications information by providing the application name or uuid |
| GET /subscriptions | Get the subscriptions meta information of an API by providing the API uuid and application uuid |

Gateway REST API v2

# Extension Points

WSO2 API Manager Gateway is highly extensible. Several extensions can be done as follows.

- Custom Message Builders and Formatters

  *Examples:*

  Engaging a new content type

  Writing a Custom Message Builder and Formatter

- Custom API Policies

  *Examples:*

  Adding a custom API Policy

  Create a Custom Policy